

Team Number: 57

Team Name: *Pulse Power Innovators*

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2.5 MW Pulse Power Supply (Modulator) — Los Alamos National Laboratory Prototype Study

Executive Summary

Group 57 — *Pulse Power Innovators* — is developing a low-power, proof-of-concept prototype that demonstrates how pulsed power systems operate in high-energy applications. The project is inspired by the 2.5 MW Pulse Power Supply (Modulator) used at Los Alamos National Laboratory, which delivers powerful, controlled pulses to drive klystrons and other high-voltage equipment.

Our goal is not to reproduce the full-scale system but to demonstrate the fundamental operation of a pulse power modulator safely and accessibly. The prototype will generate controlled voltage and current pulses at low power levels (12–24 V) to model how energy is stored and released in larger machines. This project shows that the team understands the principles of pulse formation, energy transfer, and switching control that make high-power modulators possible.

The project's scope includes simulation, design, and construction of a tunable low-voltage pulse generator. The prototype will be tested and evaluated for pulse quality, timing control, and repeatability. This demonstration will provide a clear, educational representation of pulsed power concepts while showcasing the team's design, analysis, and testing abilities.

Although the concept is not unique in the research world, the project's value lies in its educational and practical relevance. The prototype reflects the same physics and engineering that power advanced equipment used in national laboratories, aerospace systems, and medical imaging. By successfully creating and demonstrating this prototype, the team will show that they can understand, analyze, and construct systems inspired by industrial and research-grade pulsed power machines.

The customers and manufacturers of this work include Los Alamos National Laboratory and other organizations involved in high-power, high-voltage systems. These institutions can benefit from the project in two ways. First, by identifying students who have developed hands-on understanding of pulsed power systems, they gain access to a future workforce already familiar with the principles behind large-scale modulators. Second, the prototyping process itself may yield new,

innovative approaches to pulse control or energy delivery, which could offer practical improvements to existing systems. Beyond that, the project serves as an educational platform for students and researchers learning about pulse power design and testing.

The team will complete simulations, build the hardware prototype, and test the circuit to verify its operation. Results will be documented and presented for review by the project mentor, Jesus Alejandro Valladares, of Los Alamos National Laboratory.

Introduction and Problem Statement

Pulsed power delivers bursts of energy by storing charge slowly and releasing it very quickly through a controlled switching network. Unlike sinusoidal sources, pulsed systems often resemble square or triangular waveforms in their voltage/current profiles, enabling **high instantaneous power** from moderate average power sources. This operating mode is critical to applications that require short, intense energy delivery: particle accelerators (klystrons, electron guns), radar, certain medical devices, and industrial systems.

Designing, maintaining, and troubleshooting **megawatt-class modulators** is impractical in typical student laboratories due to safety, cost, and facility constraints. Yet these systems underpin technologies used in national security, aerospace, infrastructure, and advanced research. This creates a gap between what students can safely build and what industry actually runs.

Problem / Need. The field needs more engineers who understand pulsed-power behavior, safety, and measurement. Every installation is somewhat customized; requirements (voltage, current, pulse width, repetition rate, duty factor) differ across missions and facilities. Because full-scale systems are not standardized and are unsafe for teaching labs, students rarely gain hands-on experience. An **accessible, low-voltage educational prototype** closes this gap by making pulse formation, energy transfer, and waveform control **visible, measurable, and repeatable**.

Educational and Workforce Relevance. For students, the prototype turns theory into practice: it connects circuit models to observable scope traces, and timing diagrams to real switching events. For institutions such as Los Alamos National Laboratory and

similar organizations, the project identifies motivated trainees who can transition into high-reliability environments with a baseline of relevant skills and safety awareness.

What this project demonstrates. The team will demonstrate the internal energy conversion of a modulator: stored energy is shaped into controlled pulses whose **timing, width, repetition rate, and current** are governed by the design, even when operating at safe low voltages. This shows that one can perform “high-energy work” not by maintaining enormous continuous power, but by **storing modest energy and releasing it rapidly**.

Prototype Application Focus — Medical Pulse Demonstrator

To make the demonstration concrete and medically relevant, the project adopts a **TENS-style constant-current biphasic pulser** as the experimental model. The prototype will:

- Operate from **12–24 V** input for safety and accessibility
- Deliver **charge-balanced biphasic pulses** to a dummy load (e.g., **1 k Ω**) that emulates tissue impedance.
- Support **1–20 mA** output current, **50–500 μ s** pulse width, and **1–100 Hz** repetition rate.
- Provide timing control via a small MCU/timer and current regulation via an op-amp/MOSFET stage.

This choice mirrors real medical pulse delivery (timing accuracy, charge balance, current compliance) while remaining inexpensive and human-safe. It also gives clean oscilloscope waveforms for instruction and reporting.

Broader Impact. The project strengthens education-to-industry pathways. It provides a tangible platform for students to learn safe handling, measurement, and analysis of pulsed systems; helps mentors evaluate candidates on relevant skills; and may reveal incremental improvements or cost-savvy practices suitable for resource-constrained labs and field settings.

Written Report Roadmap. The remainder of this report is organized as follows. First, the Proof of Concept Approach is presented, outlining how the team will design a low-voltage prototype to model a high-power modulator. Then, Team Capabilities and Facilities is discussed, describing the members' responsibilities and their respective available resources. Afterwards, the Project Scope is laid out, and the Measurements and Success Criteria are explained. Finally, the ABET Considerations are discussed, focusing on sustainability, manufacturability, and safety, and in conclusion, a summary of the expected outcomes and supporting appendices for the budget and schedule is provided.

Customer Discovery

Proximate customer: Project mentor **Jesus A. Valladares** and collaborating faculty.

Ultimate customers: Organizations that design, operate, or maintain high-power pulsed systems (national labs, aerospace/defense contractors, specialized medical/industrial equipment makers).

Need & Value Proposition. Factories need engineers who can read schematics, simulate switching stages, verify timing, measure pulses, and document results. This project creates a low-risk pathway for training and screening such candidates. It also produces a repeatable demonstrator that can be reused for outreach and on-boarding.

Proof of Concept Approach

The team will build a low-power prototype with a pulse output of about 1 kW in order to verify the selected topology. With laboratory-safe voltage and current levels, this model will mimic the timing and switching properties of a full-scale 2.5 MW modulator. The proof of concept will concentrate on three quantifiable results: **1. Pulse Waveform Fidelity:** Produce 1 ms-wide pulses with steady repetition at 120 Hz and little droop. **2. Switching Synchronization:** Verify the timing control of many solid-state devices to deliver pulses simultaneously. **3. Thermal and Efficiency Metrics:** To evaluate scalability, measure component heating and switching losses under repeated pulsing. The prototype will have a programmed DC power source for consistent operation and a resistive dummy load to mimic klystron impedance.

Team Capabilities and Facilities

Roles (current snapshot).

- **Michael Asfaw:** Team lead for documentation, organization, meeting notes, and deliverable integration.
- **Sean Reeves:** LTSpice simulation lead (topology selection, parameter sweeps, waveform analysis).
- **Nicholas Bockrath:** data analysis and presentation support
- **Diyorbek Egamov:** test planning
- **Nicholas Pelino:** hardware assembly

Hybrid collaboration and constraints. Three members are remote (Nevada, Midwest, East Coast), two are in-person in Arizona, and the mentor's schedule varies. To manage this, the team will maintain a weekly cadence with shared notes, version-controlled schematics, and a single **build owner** for the physical prototype. If shipping is required, packaging and post-shipment validation checklists will be used to prevent damage and maintain configuration control.

Facilities & Tools. LTSpice for simulation; access to a basic electronics bench (DC supply, DMM, entry-level oscilloscope); standard hand tools and soldering. Where needed, low-cost USB logic analyzers and current-sense shunts will be employed to capture timing and pulse shape.

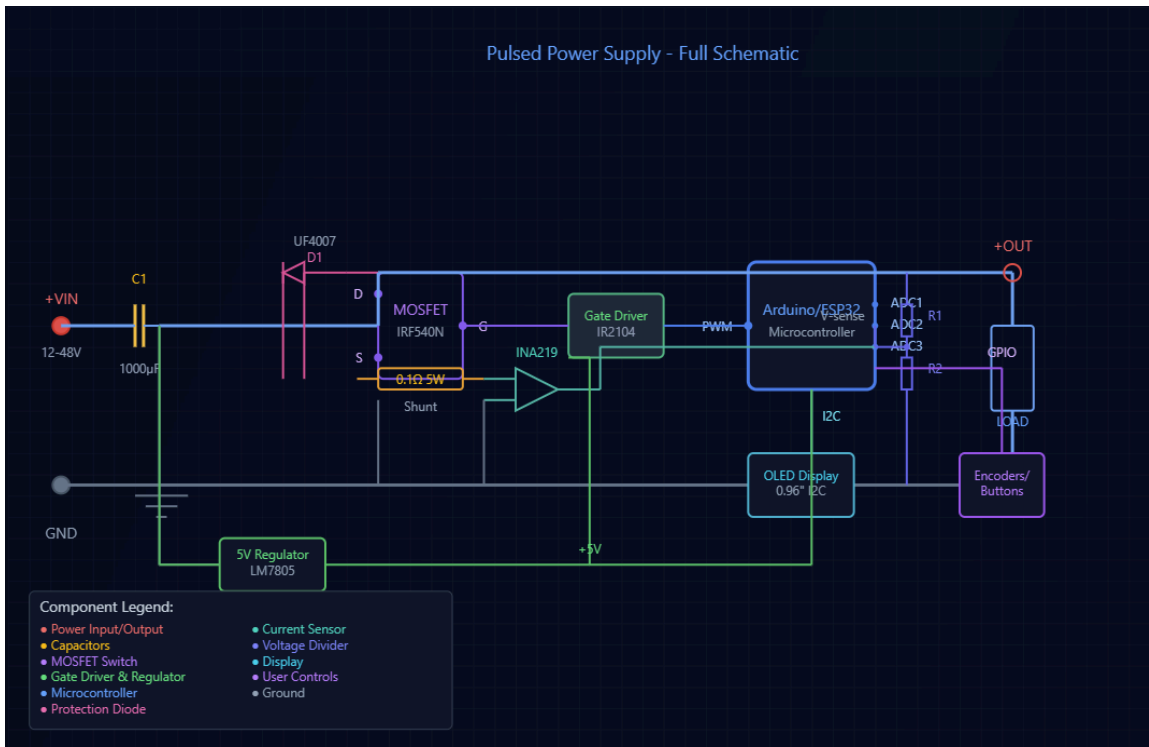
Scope

The project will deliver a **low-voltage, low-current prototype** that emulates the operating characteristics of a large modulator while remaining safe and affordable. Scope includes:

- **Simulation:** Topology selection; LTSpice models of the switch stage, current regulation, and timing; waveform validation (pulse width, rise/fall, repetition rate, charge balance).
- **Electronics design:** Op-amp current source with MOSFET regulation;

MCU/timer control; over-current and compliance limits; output enable/disable.

- **Prototype build:** Perfboard or simple PCB; labeled test nodes; connectors for dummy loads; EMI-aware wiring and grounding.
- **Measurement & verification:** Oscilloscope captures of current/voltage pulses; measurement of pulse width accuracy ($\pm 5\%$), repetition rate stability, and charge balance for biphasic operation.
- **Safety:** TVS protection, bleeder resistors, insulated connectors, and documented test procedures.



Milestone	Description	Target Completion
Simulation & Design Selection	LTspice and MATLAB modeling of PFN and switching circuits	Sep 2025
Prototype Development	Construction of low-power IGBT-based pulser	Oct 2025
Proof-of-Concept Testing	Validation of 1 ms pulse generation and timing accuracy	Nov 2025
Performance Analysis	Compare measured vs simulated efficiency and pulse shape	Dec 2025
Final Report & Demo	Summarize findings, propose full-scale design recommendations	Dec 2025

Measurements and Success Criteria

- **Timing:** Pulse width settable **50–500 μ s** with $\leq \pm 5\%$ error; **1–100 Hz** PRF with $\leq \pm 2\%$ error.

- **Current regulation:** 1–20 mA across **500 Ω –2 k Ω** loads; flat-top current within $\pm 5\%$.
- **Charge balance:** Integrated biphasic pulse pair net charge $\approx 0 \mu\text{C}$.
- **Waveform quality:** Minimal overshoot/ringing with appropriate snubbers; documented rise/fall times.
- **Traceability:** Simulation waveforms correspond to measured lab captures within reasonable tolerance.

ABET Considerations

Sustainability: Pulsed power technology is inherently linked to sustainability in the form of its energy usage. The alternative to supplying a quick pulse of voltage is through conventional DC or AC methods, which may not be necessary for all applications. Controlled bursts will **reduce power consumption**, which can be particularly useful in machinery that require extremely high-voltage levels to run, such as a particle accelerator.

Manufacturability: Given the constraints of the supplies allocated to the group, the pulsed power display will be **easier to manufacture** when compared to industrial applications. During the design process, the team is focused on making an affordable demonstration without sacrificing quality of results. This pulse power supply is not meant for high-voltage but to demonstrate to a customer how this method works and how it can be implemented in certain machinery. The team's goal is to design a power supply not only easily comprehensible to a customer, but one that is easily reproducible as well.

Safety: The specific low-voltage topology in this project was chosen with the idea of customer safety in mind. Pulsed power modulation is much more common in high-voltage applications, but due to **safety concerns** commonly present in such situations, a low-voltage model only for demonstration will be used instead. The design will also use devices previously mentioned to ensure safety while testing, such as bleeder resistors and TVS diodes.

Conclusion

This project creates a safe, low-cost platform to **observe and measure pulse formation** with professional discipline. It demonstrates team competency in simulation,

design, measurement, and documentation while aligning educational outcomes with the needs of high-power, high-voltage organizations.

Appendix A — Budget

•**Base lab-friendly build:** (Arduino/STM32 or 555; op-amp; 2× MOSFETs; sense resistor; TVS; RC snubbers; enclosure; 12–24 V brick).

•**Notes:** Reuse of existing bench supply/enclosure can reduce cost; add small contingency for shipping or spare parts.

Appendix B — Schedule

•**Weeks 1–7:** Research, literature review, mentor sync, initial LTSpice environment.

•**Weeks 8–10:** Topology down-select; parameter sweeps; schematic locked for report.

•**Weeks 11–13:** Begin hardware prototyping; bench bring-up; measurement plan.

•**Weeks 14–15:** Capture results; cross-check with simulations; finalize report & presentation.

References

[1] Mentor guidance and internal notes related to pulsed modulator operation (LANL/ASU collaboration), 2025.

[2] Representative literature on modern converter/pulsed modulators (IEEE venues).

[3] Educational resources on constant-current pulsing and charge-balanced drive for medical/biophysical systems.